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AZMI HAIDER

SUMMARY

PhD candidate in Generative AI and Computer Vision at the University of Haifa. My research focuses on developing, training, and evaluating deep generative models, with an emphasis on flow matching, diffusion models, transformers, probabilistic inference, and efficient generative sampling. My work spans image and 3D generation, continuous neural representations, and scientific machine learning.

SELECTED PUBLICATIONS

- A. Haider and D. Rosenbaum. **Direct Flow Neural Processes: Efficient Sampling via Flow Step Amortization.**
 ICML 2026, Workshop on Structured Probabilistic Inference and Generative Modeling.
- A. Haider and D. Rosenbaum. **Predicting 3D Structure by Latent Posterior Sampling.**
 ICLR 2025, Workshop on Frontiers in Probabilistic Inference: Sampling Meets Learning.
- A. Haider and D. Rosenbaum. **Tokenized Neural Fields: Structured Representations of Continuous Signals.**
 NeurIPS 2025, Workshop on Structured Probabilistic Inference and Generative Modeling.
- D. Aibinder, A. Haider, and D. Rosenbaum. **Empirical Bayes Flow Matching for Continuous Cryo-EM Heterogeneity.**
 Under submission (2026).
- N. Privman-Horesh, A. Haider, and H. Hel-Or. **Forgery Detection in 3D-Sensor Images.**
 CVPR 2018, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops, 2018, pp. 1561-1569
- A. Haider and H. Hel-Or. **What Can We Learn from Depth Camera Sensor Noise?**
 Sensors, 22(14):5448, 2022.

PROFESSIONAL EXPERIENCE

2022 – 2023	Teaching Assistant – University of Haifa, Israel Taught Image Processing and Advanced Computer Vision courses to undergraduate students, leading tutorials, project mentoring, and office hours.
2021 – 2022	Data Science Intern (Face Recognition) – Intel Corporation, Haifa, Israel Developed image preprocessing, detection, and synthetic data generation pipelines for deep-learning-based face recognition systems.
2015 – 2021	Computer Vision Internships (3D Cameras & LiDAR) – Intel Corporation, Haifa, Israel Designed evaluation methodologies for 3D camera and LiDAR quality across 2D/3D sensing pipelines used in deployed computer vision systems.

EDUCATION

2022 – PRESENT	Ph.D. in Computer Science – University of Haifa, Israel Thesis: <i>Neural Representations for Probabilistic Inference and Generative Modeling</i> . Advisor: Dr. Dan Rosenbaum . Expected graduation: November 2026.
2017 – 2020	M.Sc. in Computer Science – University of Haifa, Israel Thesis: <i>Forgery Detection in Depth Images</i> . Advisor: Prof. Hagit Hel-Or .
2012 – 2017	B.Sc. in Electrical Engineering – Technion – Israel Institute of Technology Focus: computer and software engineering.

TECHNICAL SKILLS

Programming: Python, PyTorch, OpenCV.
Generative Modeling: Diffusion models, flow matching, transformers, normalizing flows, variational autoencoders, autoregressive models.
Computer Vision: Implicit neural representations, 3D reconstruction, NeRF, Gaussian Splatting, probabilistic inference, representation learning.

HONORS AND AWARDS

2022 – 2025	Excellence Scholarship – University of Haifa
2022 – 2025	PhD Research Fellowship – Data Science Research Center, University of Haifa
2020	M.Sc. graduation with honors – University of Haifa
2011 – 2016	Full Scholarship (Outstanding Students Program) – Technion

LANGUAGES

Arabic — Native/Fluent
Hebrew — Fluent
English — Fluent
Spanish — Beginner